FACULTY OF SCIENCE
EDUCATIONAL SUPPORT
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CONCEPT NOTE

Master's theses at the Faculty of Science

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1 Introduction

The Bachelor-Master reformation has led to a number of changes in various educational programmes, for example concerning the Master's thesis. At the Faculty of Science, a thesis of 20 credits was expanded to a Master's thesis of 30 credits in most programmes. In this concept note, an overview is given of the different aspects of the Master's thesis and the Faculty's vision on these aspects. The text is based on the existing educational vision of the Faculty, enriched by conclusions from discussions within the Working Group Master's Thesis[1] in academic year 2009-2010. The different programmes are still able to add their own accents and keep a certain autonomy. The concept note applies to Master's theses of 15 credits or more.

[1] The Working Group Master's Thesis was established at the request of the Dean during the summer of 2009. Members: Iris Peeters (Chair), Bavo Meuwis (Secretary), Philippe Muchez (Vice Dean), Katrijn Dirix, Marc Hendrickx, Steven Huyghe, Johan Quaegebeur, Inge Serdons, Karen Stroobants, Patrick Van Dijck, Lucie Van Hove, Gert Verstraeten.

2 Conditions and Objectives

2.1 Conditions

A student may enroll in the master's thesis at the start of the academic year if they still need to obtain a maximum of 72 credits to earn the master's degree. If a student does not meet this requirement, they cannot begin the master's thesis.

A student who meets the 72-credit requirement at the start of the second semester is not allowed to enroll in the master's thesis at that time. The earliest they may begin the thesis is in the following academic year.

2.2 Objectives

As is the case with every programme unit, at the completion of a Master's thesis a list of learning outcomes should be reached. These learning outcomes are closely related to the intended learning outcomes of the entire study programme. Therefore, its connection to the rest of the programme is very important: which skills do students have when they start their Master's thesis? What can be expected at the start of the Master's thesis process, and which learning outcomes deserve further attention during the Master's thesis process.

For the Master's thesis, the emphasis is on the competences of students to actively contribute to scientific research. At the Faculty of Science, the following specific learning outcomes are aimed for and evaluated:

- Formulating research questions with the help of the supervisor, and performing the research.
- Acquiring information independently and assessing its relevance for answering the research questions.
- Acquiring the attitude to work on scientific research in a team (with fellow Master students, PhD students, …).
- Learning to communicate in a scientific register through collaboration with fellow students and researchers.
- Keeping track of and analysing developments in the chosen area, through a thorough training and by making contact with the current research in that area.
- Using adequate experimental or theoretical methods and techniques.
- Critically analysing the results and their interpretation.
- Reporting and presenting the original results in an orderly way and placing remaining questions in the right perspective, and linking to techniques and results from the literature as well as actual and future research.

In order to reach these learning outcomes, students should receive enough guidance without restricting their autonomy.

3 Preparation during the remainder of the programme

At the start of the Master's thesis students are expected to have gained a (thorough) basic knowledge on the thesis topic in the Bachelor programme and the first phase of the Master programme. Moreover, students already have basic competences in acquiring information independently, analysing this information, communicating on scientific findings, etc. They will further develop these competences during the Master's thesis.

Programme units from the Bachelor such as "Science Communication" contribute to the development of these competences. However, it needs to be taken into account that not every student who starts a Master programme has followed the Bachelor programme within the same discipline at KU Leuven.

Also, specialised programme units in the Master programme are a direct preparation for the Master's thesis. Furthermore, they can be a direct link to actual research. Additionally, some of the Master programmes organise a preparatory programme unit dealing with research skills and/or content-related orientation of the students. The feasibility and content of such a programme unit can be considered in each Master programme.

3.1 Examples

In the Master of **Geography** the programme unit "Research Seminar" is programmed in the second semester of the first phase of the Master. In this course, students learn how to analyse a scientific article, in order to extract the objectives and select an adequate method. Based on these experiences, students then have to formulate and analyse their own research questions in a 'start note'. Ideally, this is done in the framework of their Master's thesis, although it is still possible to change the thesis topic afterwards.

An alternative for a specialised programme unit is an internal or external internship, such as for example in the Master of **Chemistry**. Within such an internship, specific skills can be trained that are not handled in other programme units.

4 Determination of the topic

4.1 Faculty regulations

The faculty regulations describe how supervisors as well as students can propose research topics for master's theses. Guidelines can also be found regarding the publication and assignment of thesis topics.

4.2 Examples

Each programme decides which system is used to assign thesis topics to the students.

Several Master programmes of the Faculty of Science use the KU Leuven system. Within this system students express their preferences, based on which the system allocates the topics, giving as many

students as possible their first or second choice. For more information: Master's thesis: existing topic - Faculteit Wetenschappen KU Leuven.

In order to streamline the process of proposing topics by students, a good practice is to work with a standard form that students have to submit by a specific deadline. A sample form, based on that of the Master of Computer Science (Faculty of Engineering), can be found here: https://wet.kuleuven.be/studenten/masterproef/template-thesis-topic-proposal.docx.

5 Guidance of the Master's thesis

Student guidance can vary enormously between different supervisors. This is, amongst others, related to the large variety in Master's thesis types and is therefore is not a problem. However, it is important that there is **no under- or over-guidance**. On the one hand, students have the right to be guided sufficiently and get feedback on their learning process. On the other hand, students must get the opportunity to develop themselves independently.

5.1 Guiding team

The guidance for a Master's thesis is sometimes done by postdocs or PhD students and sometimes by the supervisor or by co-supervisors. For this reason, the criteria in this text are formulated for the guiding team as a whole, without making a difference between the mentor and (co)supervisor. The **supervisor,** however, carries the **final responsibility** for the guidance of a Master student and decides which tasks are delegated and to whom. In case of multiple supervisors the final responsibility is assigned to one supervisor. The faculty regulations specify who may act as (co-)supervisor of a master's thesis.

As PhD students often play an important role in the guidance of Master's theses, an information session on this topic is added to the educational programme for PhD students.

5.2 Profile of a good mentor

In order to clarify what is meant by good guidance, a profile of a good mentor can be formulated, whereby the expectations are made clear for lecturers and students.

In paragraph 5.2.1 and 5.2.2 a profile is formulated for supervisors and mentors of the Faculty of Science based on a norm of minimal guidance and maximal guidance. This profile is applicable for both (co)supervisors as well as daily mentors, whereby the supervisor keeps final responsibility.

5.2.1 Formulation of minimal guidance

A good mentor minimally fulfils the following criteria:

- The supervisor is scientifically active and up-to-date within the research domain of the Master's thesis.
- The supervisor ensures that the topic and process of the Master's thesis allow the student to reach the learning outcomes as defined by the (P)OC and/or the Faculty.
- The supervisor guards the scientific character of the Master's thesis.

- The supervisor guards the quality and feasibility of the research plan, which originates from discussion between the supervisor and the Master's student. In this research plan, the research project (topic, method, …) is formulated, and practical arrangements are made on the deadlines, practical/technical support, and guidance.
- The supervisor is responsible for freeing up enough time to have discussions with each student and to follow up their research closely.
- The supervisor:
    - helps with the planning, the development and potentially the refinement of the research;
    - teaches the student basic techniques that are not taught elsewhere in the programme;
    - helps the student to place the research in a broad context;
    - is present when the student presents their work (in seminars, in conversations with the guiding team, …) and gives feedback to the student;
    - brings the student in contact with the members of the concerning research unit(s);
    - teaches the student (if applicable) how data management is handled within the research unit and clearly communicates the expectations regarding this for the master's thesis.
    - maintains sufficient personal contact with the student to ensure that the student does not rely excessively on generative AI and remains personally responsible for the work.
    - advises the student on the process of the scientific project, encourages and challenges the student and makes the student enthusiastic;
    - decides – in dialogue with the student - on the structure of the text before the student starts writing;
    - checks the structure of and reasoning in the student's text at least once, giving feedback to the student;
    - offers the opportunity to valorise the Master's thesis (for example as a paper, an article, an abstract, a presentation, a conference contribution, …); this valorisation does not need to happen during the academic year of the thesis;
    - creates a realistic framework that allows the student to finish the Master's thesis in time;
    - evaluates the progress of the work together with the student and refines where necessary;
    - ensures that the evaluation criteria are clear and transparent to the student.
    - gives feedback during as well as after the process.

5.2.2 Formulation of maximal guidance: less guidance, more autonomy

Apart from a minimal guidance, the Faculty of Science also defines a maximal guidance, as in case of too much guidance, the student cannot be evaluated properly.

The guiding team must guard that the independence of the student is not threatened and that the student has the opportunity to demonstrate to have reached the learning outcomes themself. Over-guidance can be a stumbling block in this matter. Moreover, it is not fair towards students that received less intensive guidance. It is the responsibility of the supervisor to inform all members of the guiding team.

The following guideline should be followed:

The language of the student's text can be checked only one time at most. After this, feedback on the content can still be given. This applies to both the supervisor as well as the mentor (if there is a mentor).

The consequence of defining a maximal guidance is that not all products will be of very high quality. This may have other consequences, which, however, do not weigh up against the disadvantages of over-guidance:

- Some students may not reach the learning outcomes of the Master's thesis, despite the offered guidance, and will therefore **fail**. This is a valid option, as in the framework of the Bachelor-Master structure, a Bachelor degree is a final degree. This means that a Bachelor degree is no longer a guarantee that a Master degree will be reached, as the objectives of both degrees are different. In other words, not every Bachelor has the capabilities of reaching Master learning outcomes.
- In some cases, the results of a Master's thesis are important for the research of a PhD student, which can lead to over-guidance. This may seriously limit the learning opportunities for the Master student, and could lead to unfair situations compared to other Master students. If the difference in guidance is too large, the results may give a misrepresented image and lead to an unfair evaluation.
- Sometimes the argument is used that Master's theses are very important, in case the student applies for a PhD scholarship (FWO). However, this needs to be put in perspective. An FWO scholarship is only for the top 20% students, for whom we can assume that their Master's thesis is of high quality regardless.

5.3 Profile of a good Master student

To guarantee high-quality guidance by the guiding team, it is important to formulate what can be expected of a Master student. This is done in the profile below, which shows a general framework that can be adapted to programme-specific situations.

It is the final responsibility of the Master student to finish the Master's thesis. Additionally, it is the student's responsibility to make maximal use of the guidance that is offered by the supervisor or by the programme.

In the framework of the Master's thesis, the following can be expected of the student:

- The student ensures to be thoroughly informed regarding the expectations concerning the Master's thesis (prerequisites, general objectives, procedure, evaluation criteria).
- The student shows interest in the research domain of the Master's thesis.
- The student is partly responsible for the quality of the research plan, which originates from discussion between the supervisor and the Master's student. In this research plan, the research project (topic, method, …) is formulated and practical arrangements are made concerning the deadlines, practical/technical support, and guidance.
- The student follows the agreements that are written in the research plan, and were made in mutual agreement between student and supervisor.
- The student proactively contacts the supervisor and possible mentors to keep track of and discuss the status of the Master's thesis.

- The student is open for suggestions from the supervisor and possible mentors during the process of the Master's thesis.
- The student hands in (parts of) the Master's thesis in time as per agreed time schedule, so that the supervisor has enough time to give feedback.
- The student does not commit fraud or plagiarism. The student correctly refers to the literature.
- The student uses generative AI in an honest and responsible manner and communicates this transparently via the code of conduct and, if applicable, the contribution statement.
- The student ensures (if applicable) proper data management, in line with the agreements made with the guiding team.

In case of a duo Master's thesis, students act collegially towards each other, share information and data in line with the agreements made with the supervisor (and possible mentors), and mention their individual contributions in the thesis manuscript.

1.1 Timing of guidance

The Faculty of Science recommends that, ideally, four large feedback sessions are organised, in which the supervisor, the mentor (if there is one), and the student should be present:

- September - October: discussion of planning
- December - January (could be with oral presentation): progress
- March (could be with oral presentation): progress
- May: discussion of text

During each feedback session, the following elements should be discussed:

- Planning: based on activities instead of products, as the course of research is often unpredictable;
- Process (level of independence, problem-solving capabilities, functioning in team, level of creativity, …);
- Product (results, discussion of results, literature review, text, …).

The list of evaluation criteria can implicitly or explicitly be used during the feedback sessions (see paragraph 9). This list is communicated to the students.

It is strongly recommended that the student gives at least one presentation during the process of the Master's thesis, to be prepared for the final presentation at the defence. Optionally the readers can be involved. With this presentation, the students get the opportunity to get feedback on their presentation skills. This is an opportunity students only rarely get during their programmes.

It is important to know which skills and competences students already have at the start of their Master's thesis, especially the competences that are described in the intended learning outcomes of the Master's thesis. This way, students can be optimally prepared during their Master's programme and guided during their Master's thesis.

6 Time spent on the Master's thesis

A typical Master's thesis at the Faculty of Science accounts for 30 credits.
Some exceptions are:

- Master of Tourism: 15 credits (1-year Master)
- Master of Statistics and Data Science: 24 credits (cfr. Faculty of Engineering)

This means that a student should spend between 750 and 900 hours on a typical Master's thesis. This corresponds to an average study time of between 23 and 27 hours a week, if half of the lecture-free periods are accounted for (33 weeks).

To check whether these numbers correspond to reality, a study time measurement can be organised by a programme. This should be done by means of the "time-writing method", whereby the students keep track of the time they spend on their Master's thesis.

If a programme decides to organise a study time measurement with **time writing,** the following remarks should be taken into account:

- It cannot influence the evaluation of the student.
- It should be decided in advance whether time writing is obligatory or on a voluntary basis. In both cases, it should be kept in mind that the results are estimates and not measurements.
- It can be useful to ask the students to hand in their study times weekly or monthly.
- It should have added value for the student. For example, it can be an exercise for their later professional life, where time writing is also asked sometimes.
- Time writing can be used in the guiding process: "What have I already done?", "What do I still need to do?", … This also creates feedback for the supervisor.

It is important that students learn how to plan their time. During the Master's thesis, the student should be able to plan more independently towards the end.

The faculty believes that the standard should be to submit and defend the master's thesis during the June examination period. Intentionally postponing the thesis to the third examination period in September in order to gain extra time to work on it is not supported and should, in principle, result in a lower grade. Only in student- or discipline-specific circumstances (e.g., student illness, supervisor unavailability, failure of research equipment during the thesis process), or in case of not passing in the June period, may the September examination period be used. Supervisors play an important role in helping to ensure that defending in June is feasible for the student.

7 Form and products

7.1 Language and layout regulations

7.1.1 Language of the Master's Thesis

The master's thesis must be written in the language of the program. More detailed guidelines can be found in the faculty regulations.

7.1.2 Title- page of the Master's thesis

A **template** for the title page of the Master's thesis can be downloaded at http://wet.kuleuven.be/reglement/master-thesis, in Word as well as in LateX.

Attention! The title page should be in the language of the programme, regardless of the language of the thesis itself.

- Use the same font type, font size and font colour as used in the template.
- The font size may be smaller in relation to the length of the title.
- The cover itself has to be white.

Only in case of a **joint diploma** between institutions (for example Geography, Tourism), the information of the other institution may be placed on the title page. This is done by the programme, so that all students of this programme use the same title page.

7.1.3 First page of the Master's thesis

At the bottom of the first page, the following text should be written concerning **copyright**, in the same language as the rest of the Master's thesis:

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Zonder voorafgaande schriftelijke toestemming van zowel de promotor(en) als de auteur(s) is overnemen, kopiëren, gebruiken of realiseren van deze uitgave of gedeelten ervan verboden. Voor aanvragen tot of informatie i.v.m. het overnemen en/of gebruik en/of realisatie van gedeelten uit deze publicatie, wendt u tot de KU Leuven, Faculteit Wetenschappen, Celestijnenlaan 200H bus 2100, 3001 Leuven (Heverlee), telefoon +32 16 32 14 01.

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In the Master of Geography this only applies to students that are registered at KU Leuven.

7.1.4 Disclaimer

In the context of KU Leuven's Open Access policy, master's theses that receive a grade of 14/20 or higher are made publicly available via Limo. For this reason, each master's thesis must include a disclaimer stating that it is a student work submitted as part of an academic program. This is done by inserting a cover page at the beginning of the thesis (for example, before the foreword). You can download the cover page with the disclaimer .

7.1.5 Popularising summary

One page needs to be added to the Master's thesis with a popularising summary, which should be understandable to a broad public. This summary should be uploaded through an obligatory field in the metadata form. It is not necessary to include the popularising summary in the thesis itself.

The length of the summary is restricted to one A4 page (3500 characters) and needs to be filled in without any layout. For Master's theses in Dutch programmes the popularising summary should be in Dutch. For theses in English programmes, the language is Dutch or English.

7.1.6 Layout content

The Master's thesis needs to be consistent in relation to bibliographic and other references. An overarching list of bibliographic references is provided at the end of the thesis. Finally, references to figures, tables, appendices, and similar structures need to be consistent.

The following items are present in the Master's thesis:

- **Preface:** The preface should summarise the general aim of the work. People, offices and companies should be thanked for their support.
- **Contribution statement (depending on the programme):** This section of the master's thesis outlines the student's individual contribution and the support or input provided by others, such as the (co-)supervisor, daily advisor, fellow student, etc. This is particularly relevant in the case of a duo master's thesis. If the student has made substantial use of generative AI, this must also be clearly indicated in the contribution statement in a way that is transparent to both readers and the supervisor.
- **Summary** (max. 2 pages for each language): The summary should present the most important aims and conclusions of the dissertation in a brief text of maximum two pages, written both in Dutch and in English for a Dutch Master programme, and only in English for an English Master programme.
- **List of abbreviations and list of symbols**: These are lists of the most important abbreviations and symbols used in the work, mentioning indices, meaning and measures used.
- **Table of contents**: The table of contents should be arranged neatly and must refer to the pages of the different sections (maximum three levels). Preceding chapter one, Roman numerals (I, II, ...) should be used for the page numbering. From chapter one onwards, Western Arabic numerals (1, 2, ...) should be used.
- The **chapters** should include the following:

- A thorough introduction including the overall approach and research methods, and placing the research in a larger context;
- Results and discussion of the research;
- A conclusion including a global discussion of the research results. Suggestions for future work may be added here.

- **Appendices**: The appendices should include parts of the research that are essential for the work but may hamper the readability of the text, e.g. because of their length (mathematical deductions, raw experimental data, examples, figures, etc.).

As **font type,** a type from the family of Helvetica Neue or Arial is used, which is prescribed by the KU Leuven. For those working in LateX, the template for PhD's of the Arenberg Doctoral School can be used: https://set.kuleuven.be/intranet-associatie/ADS/guidelines_PhDthesis.htm.

7.1.7 References and plagiarism

It is important that each programme provides **clear guidelines** to the students concerning references and plagiarism, which should be applied in all programme units. In case of the Master's thesis, good guidance can help prevent plagiarism. If plagiarism is detected despite all efforts, this needs to be discussed with the ombuds and with the Chair of the Examination Committee. More information on the university-wide vision on plagiarism can be found at http://www.kuleuven.be/plagiarism.

The policy of the Faculty of Science on scientific integrity can be read at http://wet.kuleuven.be/sbook/wi/en. The Faculty wants students to obtain a certificate in information skills during the course of their Bachelor's and/or Master's programme based on an online tutorial. **Having the certificate is a prerequisite for submitting the Master's thesis.**

TurnitIn is used as a means to detect plagiarism; for a manual go to https://wiki.associatie.kuleuven.be/toledopedia/index.php/Turnitin_Assignment.

In the context of scientific integrity, it is also important that students give sufficient **credit to other contributors** in their Master's thesis. This can be done in a separate 'contribution statement' section (see 7.1.6). An alternative approach is to address this throughout the Master's thesis text, wherever relevant.

When writing the master's thesis, students are allowed to use GenAI tools, provided that they remain fully responsible for the work they submit and that the work clearly allows for an assessment of the extent to which the intended learning outcomes have been achieved.

- Students can refer to the KU Leuven guidelines on the responsible use of GenAI.
- Students must complete the code of conduct truthfully. Just like the information literacy certificate, uploading the code of conduct is a requirement for submitting the master's thesis:

7.1.8 Page count

Every programme should give a **directive** to the students on the optimal page count of a Master's thesis. This does not need to be a binding restriction.

7.2 Oral defence

At the end of the Master's thesis the work is defended in front of a **Defence Committee**, of which at least the supervisor, co-supervisor (if there is one), and readers are members. A **Chair** needs to be appointed to take part in different defences to ensure the consistency in evaluation (see 9.3). This can be the Chair of the Examination Committee or a delegate. The defences are **organised** by the **coordinator of the Master's thesis**, who can delegate this task.

At the start of the defence, the student gets the opportunity of presenting their work. Next, the jury asks questions that can lead to a discussion. The (P)OC needs to agree on the duration of the presentation and the time spent on questions beforehand, in order to give all students equal chances. As a guideline, we can say that each student has a right to **at least 20 minutes** for the total defence.

If the reader has not had an oral discussion with the student about the work prior to the defense, they ensure during the discussion that they gain a sense of the student's critical reflection on their own work.

In the case of theses under **embargo**, it is not permitted to make recordings or photographs during the defence. With regard to the personnel present, a confidentiality obligation applies that is imposed by the employment regulations, even if there is no embargo on the thesis. There are no automatically applicable enforceable provisions for students and third parties. If they are nevertheless admitted to the defence, they may be asked to sign a confidentiality statement (see appendices).

7.2.1 Example

In the Master of Mathematics two oral defences are planned:

- A critical defence for the supervisor, readers and research group members of one hour;
- A public defence for fellow students that is popularised with regard to the content.

8 Submission of the Master's thesis

8.1 Electronic submission

All master's theses at KU Leuven must be submitted electronically via KU Loket. The purpose of this system is to store the theses in a central database and make them accessible through the library system.
The faculty regulations contain guidelines for submitting the master's thesis. The exact deadlines set by each program committee ((P)OC) must be communicated to students clearly and in a timely manner. This is the responsibility of the master's thesis coordinator.

8.2 Embargo

Some master's theses are subject to embargo in terms of content (e.g. in the case of patents, data covered by a confidentiality agreement, etc.). At the student's motivated request or on the advice of the supervisor, and with the approval of the master's thesis coordinator, the master's thesis can be placed

under embargo for a certain period. To this end, the student must submit the request for embargo via an online form with the approval of the supervisor and the master's thesis coordinator as an attachment: https://wet.kuleuven.be/formulieren/embargo.

There are two possible forms of embargo:

- Full embargo: Embargo on publication on the internet and intranet of KU Leuven (master's thesis will neither be publicly available nor consultable for persons with access to the KU Leuven network via the library catalogue Limo)
- Partial embargo: Embargo on publication on the internet (master's thesis will not be publicly available, but will be consultable for persons with access to the KU Leuven network via the library catalogue Limo)

Usually, the reasons why an embargo is requested are of such a nature (e.g. use of personal data, confidential data, contractual agreements) that a full embargo must be requested for both publication on the internet and intranet.

9 Evaluation of the Master's thesis

9.1 Faculty regulations

The faculty regulations outline the formal guidelines regarding the evaluation of the master's thesis.

Four separate scores are awarded for the thesis: one by the supervisor, one by each of the two readers, and one for the defense by the evaluation committee. The current weighting of these three evaluator groups in the final score is 50/15/15/20 (supervisor, two readers, defense committee). In exceptional cases, this distribution may be adjusted depending on the evaluation method used within the program. (For example, the relative weighting for the master's thesis in Geography is 8:4:4:4, while in the Master of Space Studies it is 8:3:3:6.). Nevertheless, it remains essential that the process, the written thesis, and the defense are all evaluated. In this evaluation, the supervisor and external readers play a crucial role.

9.2 Readers

The two readers assess the content and form of the written final product with an external perspective, prior to the defense. If both readers base their evaluation solely on the written document, the score becomes particularly vulnerable to fraud. Therefore, it is crucial that the readers also evaluate the student's critical insight and individual contribution through an oral interaction. This can be done, for example, via an individual conversation or during the defense, depending on the program's preference. If the reader only has the opportunity to speak with the student during the defense, it may, in exceptional cases and with proper justification, be appropriate to adjust the reader's score after the defense.

1.2 Assessment Committee

The Assessment Committee consists of the supervisor, two readers and a Defence Committee, of which the supervisor and reader(s) are also members (see 7.2). Good communication between the supervisor and the daily mentor(s) is essential to get a clear image of the process.

It is important that the supervisor informs the readers and other members of the Assessment Committee about the followed procedure. It is also an option to give the readers a document with information on the process of the concerning student; this is for example done in the Master of Geology.

The assessment of the defence is done by the Defence Committee. Involved persons that cannot be present at the defence, evidently cannot take part in the assessment.

9.3 Aiming for an objective and transparent evaluation

It is important that the evaluation of the Master's thesis is done in an objective and transparent way. Therefore, it is recommended that unambiguous criteria are used by every assessor. To achieve this, a Faculty **evaluation rubric** is developed, in which evaluation criteria for the written report, the process and the oral defence are determined based on the learning outcomes of the Master's thesis. In this rubric, the examiners can indicate to which extent the student fulfils the criteria on a scale from insufficient to excellent. If a criterion is not applicable for the concerning thesis, this can be indicated.

In the evaluation rubric, a clear distinction is made between **process** and **product**. These aspects are assessed by the supervisor (and mentors that may be involved) and the readers, respectively, as a reader cannot evaluate the process as well as the supervisor (or mentor) can.

The evaluation rubric can facilitate the discussion between different members of the Assessment Committee and can be used to give structured feedback to the student. It is not an absolute measure but an **aid** for the examiner. Therefore, no weights are coupled to the different criteria. However, the final appreciation lies between the minimum and maximum appreciation for each rubric. Further explanation can be given in the global motivation. The rubrics must be completed thoughtfully, keeping in mind that they may be requested as evidence to support scores in the event of an appeal.

The students are informed about the evaluation criteria at the start of the Master's thesis and during the guidance, the criteria can be used to give feedback. This is essential to stimulate the learning process of the student, especially since the Master's thesis is an important part of the programme in which the acquisition of essential competences is achieved. After the final assessment, the evaluation criteria can be used as a guide to give oral feedback to the students to give them the opportunity to learn.

Completed evaluation rubrics are not routinely shared with students. They are considered internal communication between evaluators and, from that perspective, do not fall under public access regulations. If a student requests access to the completed rubrics after having already received oral feedback, it is up to the master's thesis coordinator to decide whether or not to grant access. To avoid a formal appeal, it may be decided to share the rubrics with the student. However, due to the involvement of multiple evaluators, it may be preferable not to share the rubrics directly. In such cases, a clean synthesis document may be provided to the student instead.

Important note: the master's thesis coordinator may refuse to share the completed grids, but if the matter leads to a formal appeal, the appeal body may request the completed grids as evidence.

Next to the evaluation rubric, there is also an **appreciation scale**. This scale can help translate appreciations from the rubric to a mark without using mathematical calculations.

The evaluation rubric and appreciation scale can be downloaded at https://wet.kuleuven.be/onderwijs/mastersthesis/evaluation.

Finally, it is important that a neutral person attends to the different **oral defences** to ensure consistency of evaluation (the Chair of the Defence Committee). If it is impossible for one person to attend all the defences, several persons can attend different blocks of defences. Furthermore, the level of the **questions** at the defence need to be guarded by the Chair of the Defence Committee. The students' general insight as well as their detail knowledge needs to be tested. The members of the Defence Committee need to be informed correctly. One must bear in mind that the defence is essentially an exam, which comprises 20% of the final mark for the Master's thesis!

To pass the master's thesis, the student must obtain a passing grade for each of the following components: the supervisor's evaluation, the average of the two readers' evaluations, and the defense. If the student fails one or more of these components, the maximum final grade is 9/20. In the case of a non-tolerable fail (i.e. ≤ 7/20) from one of the readers and a score > 11/20 from the other reader, both readers must engage in a discussion to determine whether the student passes this component and justify their decision. If no consensus is reached regarding the pass/fail status or a possible adjusted score, the original reader scores are maintained, but the student is considered to have passed this component of the evaluation.

9.4 Submission of marks

The following procedure is followed:

- The coordinator of the Master's thesis receives a student list from the Faculty, on which all marks (also partial marks of the supervisors, readers and Defence Committees) can be filled in. The coordinator delivers this list to the Chair(s) of the Defence Committee(s).
- The marks are collected at the defence after the discussion by the Chair of the Defence Committee, who in turn delivers them to the coordinator of the Master's thesis.
- The completed evaluation rubrics of the readers and supervisors are delivered to the Chair of the Defence Committee at the latest at the defence. The Chair delivers these forms to the coordinator of the Master's thesis together with the marks. Optionally, the readers and supervisors keep a copy of their own rubric in order to give feedback to the student.
- For partial marks as well as the final mark, one decimal is allowed.
- The coordinator of the Master's thesis is then responsible for the delivery of the completed list to the Faculty. This is done according to the 50/15/15/20 system (Master of Geography: 40/20/20/20, Master of Space Studies: 40/15/15/30) and via the quotation form provided by the Faculty administration.
- The Faculty imports the marks into SAP.

9.5 Not passing the Master's thesis (failing mark or NA)

The faculty regulations describe what happens if a student does not pass the master's thesis after the second examination period, after the third examination period, and after a second enrollment.

10 Duo master's theses

A duo Master's thesis is a Master's thesis carried out by two students. The students can be enrolled in the **same programme** or in **different programmes** of the Faculty of Science or another faculty. Each student follows the Master's thesis regulations that apply within their own programme/faculty.

A duo Master's thesis can take **different forms**. However, any duo Master's thesis has to meet the following **two criteria**:

- There should be **sufficient collaboration** between the two students during the Master's thesis process, in all cases for answering a **shared research question** and optionally also for joint data collection;
- There should be **clear individual input** from each student, for example under the form of their own research question and/or individual data collection.

If these criteria are not met, there is no duo Master's thesis. Students can nevertheless still work together, for example to realise a more efficient Master's thesis supervision in the research group(s). The students then need to mention the contribution of others in their Master's thesis manuscript, for example in a separate chapter 'contribution statement' (see also 7.1.7 and 7.1.8).

Three examples of duo Master's theses that meet the above criteria are described below for inspiration:

1. A duo Master's thesis in which two students try to answer a shared research question from the **same perspective**. The two students **collect data together**. To avoid the Master's theses being limited to only reporting and interpreting the same data, each student additionally has an **own research question** that is investigated and reported on individually.
2. A duo Master's thesis in which two students work on the same topic but from distinctly **different perspectives**, for example; two students from different programmes or from the same programme but active in different research groups, each investigating the same topic from within their own (sub)discipline or specialisation. The two students **collect data separately** and then **integrate each other's conclusions** to answer a complex research question together in an interdisciplinary way.
3. A duo Master's thesis in which students **collect data together** but with the aim of answering their **own research question**. In this case, an additional **shared research question** is required to qualify as a duo Master's thesis.

Some aspects of the Master's thesis that have already been discussed higher up in this concept note, but that require further clarification in the case of duo Master's theses, are discussed below.

10.1 Determination of the topic of duo Master's theses

As mentioned above, each programme decides for itself which system to use to assign Master's thesis topics to students (see 4.2). In addition, **programmes** also **decide** if supervisors and students can propose topics for duo Master's theses. The large student numbers in some Master's programmes may be a reason to engage in duo Master's theses. For certain research topics, it can also be an added value to work in pairs (e.g. because of the size of the research topic and/or the different perspectives from which the topic can be looked at).

If a programme wants to offer duo Master's thesis topics, it needs to determine **how to implement this in the system that is used to assign Master's thesis topics to students** (e.g., how will it be made clear which topics are suitable for one and/or two students?).

In addition, the programme should decide **by whom and how exactly the duos will be formed**. For example, the programme may choose to form the duos itself after students have entered their choice of a Master's thesis topic into the system. Another possibility is that students themselves propose a joint Master's thesis topic. In that case, it should be clarified how students should make themselves known as a 'team', how agreements should be made with the supervisor(s) and how this may or may not run simultaneously with the selection of a Master's thesis topic in the system.

In the Master of Statistics and Data Science the following procedure is used:

- Students first form pairs on their own. A Toledo discussion forum allows them to find a partner.
- Via Tolinto, the student pairs make themselves known.
- Within each pair, one of the students subscribes to a Master's thesis topic in the KU Leuven system.

For the distribution of duo Master's theses on which students from different programmes/faculties collaborate, it is recommended to **work outside the systems**, as is done for topics proposed by students themselves.

Finally, a duo Master's thesis cannot be started if it is clear from the start that the two students cannot **defend in the same examination period**.

10.2 Guidance of a duo Master's thesis

The guiding team includes at least one supervisor, possibly assisted by one or more daily mentors (e.g., one per student). In the case of a duo Master's thesis involving two students from different programmes or research groups, it may be appropriate to work with two supervisors, i.e. one from each programme or research group.

A duo Master's thesis requires more coordination. It may therefore be useful for the guiding team to discuss planning with the two students at the start of the Master's thesis process (see also 5.5). The perspective from which each student will investigate the Master's thesis topic can then be determined, as well as the concrete division of work between the two students. This will ensure that each student makes a substantial contribution to the Master's thesis research and that there is also sufficient collaboration between the two students. As soon as possible in the Master's thesis process, it is made concrete how the written paper will be structured for each student (see 10.4).

Throughout the Master's thesis process, it is desirable that the two students have joint meetings with the guiding team at regular intervals to discuss progress. The guiding team can gain insight into how the collaboration between the students is going and what contribution each student has made to the research. On the one hand, this should allow for timely adjustments where necessary. On the other hand, this also enables the guiding team to adequately assess the criterion 'process' at the end of the Master's thesis for each student individually (see 10.6).

If in the course of the Master's thesis process problems arise in the collaboration between the students (e.g. a student gives up or the two students do not manage to work together constructively) it is recommended in the first place to check with the guiding team whether these problems can be solved so that the collaboration can be continued. However, if this turns out to be impossible, the Master's

ombuds should be contacted. Together with the various parties involved (students and guiding team), the Master ombuds looks at whether and how the students can complete their theses without further collaboration. The expectations towards the two students are adjusted if they each complete their Master's thesis individually.

10.3 Time spent on a duo Master's thesis

A typical Master's thesis at the Faculty of Science accounts for 30 credits. This means that between 750 and 900 hours should be spent on a typical Master's thesis by the student (see also 6.1). If the Master's thesis is conducted by a group of two students, each student is expected to invest this amount of time.

In addition to the time spent by students on the duo Master's thesis, there is also the time spent by the guiding team. There is a potential time gain here, compared to supervising two individual Master's theses, provided that the duo Master's thesis is thoughtfully designed and well organised. For example, joint meetings with the two students can ensure that certain issues only need to be explained once by the guiding team. Students can also be encouraged to help each other out first before seeking help or advice from the guiding team. If it is agreed that students write clearly delineated parts of the thesis together, time savings can also be made in proofreading.

10.4 Form and products for duo Master's theses

In the case of a duo Master's thesis:

- each student either submits an **own Master's thesis manuscript**,
- or the two students submit a **joint Master's thesis manuscript** in which both jointly and individually written texts are bundled.

In both cases, it should be clearly stated what everyone's contribution was to the various parts of the Master's thesis (see also 10.4.1).

10.4.1 Language and layout regulations

The **title page** of the Master's thesis includes the names of both students in the case of a joint manuscript. If the students are writing their Master's thesis separately, each student mentions the name and programme of the other student, in addition to their own name and programme on the title page (for example via the sentence "in collaboration with (name of student) from the Master of (name of programme)").

When students submit a joint thesis as part of a duo master's project, the title must be identical. This is to avoid issues during the plagiarism screening process using Turnitin.

At the **beginning of each chapter**, the student mentions who contributed to the chapter (the student alone, together with the other student or certain parts alone and other parts together). It is important to

make clear to the reader what the contribution has been of each student to the Master's thesis work. Copyright belongs to every person that contributed to a chapter.

As indicated above, a duo Master's thesis can take several forms. This means that the Master's thesis manuscript can also take different forms. However, every duo Master's thesis needs to include a **reasonable proportio**n that the students write **independently** of each other and that should allow both students to be evaluated independently of each other.

We give **some examples** of possible forms the Master's thesis manuscript could take:

- In the case of a duo Master's thesis on which two students each work from a distinctly different perspective, each student's written paper could consist of:
    - A (partially) joint:
        - introduction in which the central research question is framed from the two different perspectives;
        - conclusion in which the two students integrate each other's insights to answer a complex problem.
    - An individual description of the material, methods and results, as well as their own summary, which may include references to the other student's work where relevant.
- In a duo Master's thesis, two students can also work together on the same research question, collect data together and interpret these data together, but still choose to write their Master's thesis completely separately. In that case, it is important that they sufficiently refer to each other and acknowledge each other's contribution to the research. Each student then also has an additional, individual research question to address (see introduction on duo Master's theses above).
- In the Master of Statistics and Data Science, duo Master's theses have been used since academic year '21-'22. The Master's thesis manuscript of each student consists of:
    - A joint part that the two students write together including:
        - An extensive introduction regarding the central research question.
        - The methods and results of the jointly conducted research.
    - An individual part that the students write independently and which consists of:
        - A short chapter of at least five pages on an investigation that the student conducted separately (their own individual analysis, an additional experiment, etc.). To facilitate evaluation for supervisors and readers, this chapter forms a separate entity in the Master's thesis, with an own introduction, description of materials and methods, results and conclusions.
        - The summary
        - The conclusion

10.5 Evaluation of duo Master's theses

For a duo Master's thesis, the **same evaluation criteria** apply to students in the Faculty of Science as for an individual Master's thesis (https://wet.kuleuven.be/onderwijs/mastersthesis/evaluation). The evaluation rubric is completed separately for each student. Students affiliated with another faculty follow the Master's thesis regulations of their own faculty and are therefore assessed on the basis of the criteria used at that faculty.

As far as possible, it is recommended that the two students are evaluated by the **same Assessment Committee**. This implies that the same readers evaluate the two Master's theses or the joint Master's thesis.

Each student **defends** the Master's thesis **individually**, demonstrating mastery of the topic and the research conducted. If possible, the two students defend their Master's thesis before the **same committee**, **right after each other** and do not attend each other's defence (if necessary, one student is asked to leave the room when the other student has to defend). The Defence Committee tries to ask each student **more or less the same questions** about the joint part in the Master's thesis. In the case of a duo Master's thesis involving two supervisors, ideally both should be a member of the Defence Committee.

As for an individual Master's thesis, **four marks** are given to each student, one from the supervisor, one from each of the readers and one for the defence. The relative weight of these four marks is 10:3:3:4 (i.e. supervisor: 50%, reader 1: 15%, reader 2: 15%, defence: 20%), except in the Master of Geography (interuniversity programme) where it is 8:4:4:4 and in the Master of Space Studies where it is 8:3:3:6. If the situation allows, the marks are given after the two students have defended their Master's thesis consecutively. In case two supervisors are involved, they try to reach a joint mark by mutual agreement.

To pass the Master's thesis, the student must obtain (as for a 'regular' Master's thesis) a score of **at least 10/20** for the mark given by the supervisor(s), the mark of the readers (i.e. the average of the marks of the two readers) and for the defence. If one or more of these marks is lower than 10, the Master's thesis will be considered unsuccessful and the total mark will be a maximum of 9/20.

The two students will **not necessarily** receive the **same mark** for their Master's thesis. The final mark depends on their contribution to the process, their manuscript (joint part and individual part), their presentation and how they answer questions during the oral defence.

In case of **failure** to pass a duo Master's thesis (failing mark or NA), the same rules apply as described under 9.6. If only one of the two students does not pass the duo Master's thesis, the Assessment Committee and the Examination Committee must ensure that this can be thoroughly justified in case of an appeal. The Examination Committee also needs to decide (based on the advice of the Master's thesis coordinator and supervisor) whether the student can continue to work individually on the same Master's thesis topic or needs to choose a completely new topic.

Appendices to the concept note

Confidentiality declaration

This unilateral confidentiality declaration is issued to

KATHOLIEKE UNIVERSITEIT LEUVEN, Oude Markt 13, 3000 Leuven, Belgium

(hereinafter referred to as "**KU Leuven**")

by:

the **undersigned**

name:……

address: ……..

As a person attending the defence on ..... (date) of the masters' thesis, internship report, project work (delete where inapplicable) entitled
…….....……
……
of (details of the student(s))
name:
……..……
……
the undersigned will receive confidential information of (details of the organisation/company)
name: ……..……

(hereinafter referred to as "the **Organisation**")

With respect to all information and data of the Organisaton disclosed to him/her orally, in writing or in any other form, during this defence, the undersigned undertakes to

- keep them secret and confidential as long as they are not public

- treat them with due care to prevent any disclosure to third parties

- not to use them

- and not to disclose them, either directly or indirectly, to third parties.

The obligation of confidentiality shall remain in force for an unlimited period.

The obligation of confidentiality does not apply to information which
(a) was already publicly known at the time of its disclosure or becomes publicly available during the confidentiality period;
(b) was already in the possession of the undersigned before its disclosure by the KU Leuven and/or the Students;
(c) the undersigned has obtained from a third party which has no obligation of confidentiality;
(d) the undersigned has developed independently without using the confidential information of the Organisation;
(e) must be disclosed by law or as a result of a court decision, provided that the receiving Party notifies the KU Leuven.

This declaration will be governed by and interpreted in accordance with Belgian law, with the exception of the provisions on choice of law.

Any dispute relating to this declaration will be subject to the exclusive jurisdiction of the courts of Leuven, Belgium.

This declaration can be signed handwritten or with an electronic signature by means of generally accepted and available software and/or a scan or PDF of the original signature.

The KU Leuven will receive a copy of this declaration. A copy will be provided to the Organisation upon request.

Name	**First name(s)**, **Surname**
Signature	For agreement,
Date	DD month YYYY

Confidentiality declaration of external jury member

This unilateral confidentiality declaration is issued to

KATHOLIEKE UNIVERSITEIT LEUVEN, Oude Markt 13, 3000 Leuven, Belgium

(hereinafter referred to as "**KU Leuven**")

by:

the **undersigned**

name:……

address: ……..

As a member of the jury for the defence on ..... (date) of the masters' thesis, internship report, project work (delete where inapplicable) entitled
…….....……
……
of (details of the student(s))
name:
…….……
……
the undersigned will receive confidential information of (details of the organisation/company)
name: ……..……

(hereinafter referred to as "the **Organisation**")

With respect to all information and data of the Organisaton disclosed to him/her orally, in writing or in any other form, during or in preparation for this defence, the undersigned undertakes to

- keep them secret and confidential as long as they are not public

- treat them with due care to prevent any disclosure to third parties

- use them only for the performance of his/her function as a member of the jury

- and not to disclose them, either directly or indirectly, to third parties, with the exception of the other members of the jury.

Upon request of the KU Leuven or upon termination of his/her function as a member of the jury, the undersigned will immediately delete the confidential information and any copies made and the undersigned will not further use or disclose it.

The obligation of confidentiality shall remain in force for an unlimited period after termination of his/her function as a member of the jury

The obligation of confidentiality does not apply to information which

(a) was already publicly known at the time of its disclosure or becomes publicly available during the confidentiality period;
(b) was already in the possession of the undersigned before its disclosure by the KU Leuven and/or the Students;
(c) the undersigned has obtained from a third party which has no obligation of confidentiality;
(d) the undersigned has developed independently without using the confidential information of the Organisation;
(e) must be disclosed by law or as a result of a court decision, provided that the receiving Party notifies the KU Leuven.

This declaration will be governed by and interpreted in accordance with Belgian law, with the exception of the provisions on choice of law.

Any dispute relating to this declaration will be subject to the exclusive jurisdiction of the courts of Leuven, Belgium.

This declaration can be signed handwritten or with an electronic signature by means of generally accepted and available software and/or a scan or PDF of the original signature.

The KU Leuven will receive a copy of this declaration. A copy will be provided to the Organisation upon request.

Name	**First name(s)**, **Surname**
Signature	For agreement,
Date	DD month YYYY